

Anuj Tanwar

LEAD PROTOCOL ENGINEER: PERPETUALS, ORACLES AND LIQUIDATION INFRASTRUCTURE

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Remote (global) · open to relocation · IST, flexible to US / EU / Asia hours

Engineer for the parts of a trading venue that hold and move money. Until Jul 2026, Lead Protocol Engineer at Adrena, a Solana perpetuals exchange (wound down 2026): owned the Virtual Funding Rate engine for its synthetic markets, the check that several providers agree before a price is used, and the services that closed underwater positions and filled orders, while ~\$890M traded across 941 traders. Part-time at Autonom alongside, brought Treasury-bill rates and US equities into the signed feed Adrena's oracle consumed. Earlier: two years running a price oracle solo on a test network, and core contracts of an EVM perpetual-positions DEX. Rust, TypeScript, Solidity; available immediately for a team building a financial product, crypto or traditional.

CORE SKILLS

Trading mechanics: pool-to-peer perpetuals (no order book), margin, funding rates, liquidations, stop-loss, take-profit and limit execution

Market data and oracles: multi-provider price consensus, signed price publishing, TradFi feeds (rates, indices, equities, currency conversion), market-hours gating

Custody and settlement: non-custodial vaults, claims anyone can trigger, share-token NAV accounting, upgrading live programs and accounts in place, timelocked admin handover

Stack: Rust (Solana), Solidity (EVM), TypeScript, Node.js, Python (backtesting), PostgreSQL

Operations and correctness: 24/7 execution services, the venue's own Solana RPC node, verifiable builds, fuzz and invariant testing

EXPERIENCE

WhiteKnight / fatsats.io · Co-founder, Protocol & Backend

Remote · Aug 2026 - Present

Non-custodial Solana service operating positions in a third-party on-chain mining game with depositors' USDC, under fee caps each depositor sets.

- Built the custody program, live on mainnet: only depositors can withdraw their USDC, fees are capped on-chain, and claims anyone can trigger so a vanished operator cannot strand winnings.
- Built the operator service settling every ~60-second round, with a 15-minute watchdog checking vault solvency to the satoshi and an hourly audit that reconciles books against the chain.
- Tests run against the game's real mainnet code rather than a mock; the program has been upgraded in place since launch with a byte-identical wire format, so live clients kept working.

Diamond Pools · Co-founder, Contracts & Backend

Remote · Jul 2026 - Present

Quantified yield vaults on Solana around a separate on-chain mining game; pre-launch.

- Sole author of the vault program: NAV-priced share tokens in three risk tiers, payouts that halt on any failed check, tested against a simulated share-price inflation attack (early depositor diluting later ones).
- Built the strategy engine (positions sized only when expected value is positive, halts on drawdown) and the backtesting stack; found the return model overstated yield 3-4x by reading the mining program's source.

Adrena · Lead Protocol Engineer

Remote · Dec 2025 - Jul 2026

Solana perpetuals exchange, \$9.7B volume and 10,000+ liquidations lifetime; team of five; wound down 2026.

- Designed and shipped the Virtual Funding Rate engine for the exchange's synthetic markets: open-interest imbalance drives funding from the crowded side to the other, capped per market, settled over a position's life.
- Lead author of the oracle state module, where a price counted once several providers agreed, and of the publisher signing and submitting ~94,000 price batches a day under a 7-second freshness limit.
- Sole author of the services that closed underwater positions and filled stop-loss, take-profit and limit orders; ran them 24/7 with weekly maintenance windows.
- Second-largest contributor to the event-driven executor that liquidated and filled stop-loss orders the moment an account changed on-chain; ran the exchange's Solana RPC node (non-voting): ~80 requests a second at 99.88% success.
- Owned a third-party audit remediation: triaged 33 findings, patched every applicable critical, high and medium, added a 48-hour timelock on admin handover; releases were verifiable builds approved by DAO vote.

- Built gold, silver, WTI crude, S&P 500 and NVDA perpetuals, including its first US-equity markets, with market-hours attestations so nothing traded while the underlying market was closed; mainnet Apr-May 2026, pilot scale.

Autonom · Backend Engineer, Oracle & Market-Data (Rust), part-time

Remote · Mar 2026 - Jul 2026

Signed price-data provider upstream of Adrena's oracle.

- Added TradFi instruments to the production signed price feed: US Treasury-bill rates, the VIX, Nikkei 225 with JPY-to-USD conversion, and US- and Hong Kong-listed equities.
- Added a Tokyo (JPX) venue calendar and per-feed trading windows to the signed payloads, so the exchange knows when each instrument may trade.

Doot Foundation · Founder & Sole Engineer

Remote · Oct 2023 - Dec 2025

Price oracle for the Mina blockchain on zero-knowledge smart contracts; test network only, shut down Dec 2025.

- Built and ran it for two years: 20+ providers aggregated into signed prices every 10 minutes, every provider response signed before it was counted.
- Reimplemented Mina's signature verification from scratch in Solidity and deployed it on Arbitrum One, so Ethereum-side contracts can verify Mina-signed prices.
- Also primary author of the signed multi-provider price feed for Fizk (formerly zkUSD), a Mina stablecoin (Mar-Apr 2025).

Elemental Blockchain · Smart-Contract & Full-Stack Engineer (contract)

Remote · Mar 2023 - Feb 2024

Contract work on EVM trading and escrow products under Brian Lenertz, later head of Adrena and Autonom.

- Authored the Trading and Positions contracts and the trading interface for POP! Protocol, an EVM perpetual-positions DEX where orders are signed off-chain by a sequencer and settled on-chain; testnet.
- Lead contributor to EBSS, a B2B invoicing and escrow platform on Polygon: front end, its data index, and escrow-contract extensions that turn each invoice into a claim token.

SELECTED PROJECTS

- **Casper Draw:** Provably-fair lottery live on Casper mainnet: sole author of contracts, backend, payout odds, house economics (rake, pool seeding, holdback) and commit-reveal randomness with entropy committed before play.
- **pErp-man:** Commodity perpetuals DEX (gold, silver, oil, platinum) on CreditCoin testnet, 2026: top contributor; upgradeable contracts with on-chain DAO ownership handoff and a settlement service.
- **Bonding-curve launchpad:** Token-sale contracts on Arbitrum testnet (solo, Sep 2026): no admin path can move reserves; finished sales migrate liquidity to Uniswap V2 with pool tokens burned; 10 review findings closed, one critical.
- **Elekind / Gaj Dastak (Celeron IT Solutions, co-founder):** Edge-AI elephant detection and deterrence unit deployed with the Chhattisgarh Forest Department (2026): device software, thermal detection models trained on a dataset collected and annotated by hand, alerting and deterrence pipeline.
- **SearchLeads:** Stripe billing for a live B2B SaaS (2025): credit and pay-as-you-go plans, proration, idempotent webhooks so a retried event is never applied twice.

EDUCATION & TRAINING

- B.Sc. Physical Sciences with Computer Science, Hansraj College, University of Delhi, 2019-2022, CGPA 8.9/10
- Encode Club Zero-Knowledge Bootcamp, Q4 2023; Mina Navigators / zkIgnite Cohort 2 grant program